Barrel and cover's bolted design

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Index

1. Purpose & Scope 3
2. Abbreviations 3
3. General procedure: description 3
4. Calculation method: bolted joint. 4
4.1. Bolted joint description 4
4.2. Fastener stiffness 4
4.3. Barrel and cover stiffness 5
4.4. Tightening load calculation 5
4.5. Thermal load calculation 6
4.6. Internal pressure load calculation 6
4.7. Lead screw thickness verification 8
4.8. Gasket load 8
5. Calculation method: cover design 10
5.1. General approach 10
5.2. Approach as per ASME-BPVC Sect. VIII Div. 1 14
6. Calculation method: barrel thickness 24
7. Acceptability criteria 25
7.1. Bolted joint verification 25
7.2. Discharge cover verification 25
7.3. Barrel thickness verification 26
8. References 26
9. Appendix 27
9.1. Calculation Tool 27
9.2. Case study – Job P30640.002 / Item 20157 29

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1. Purpose & Scope

This procedure describes how to carry out the mechanical design of the bolted joint between the barrel casing and the discharge cover in horizontal multi-stage pumps (BB5 type). The proposed procedure does not include any verification of gasket sealing effectiveness against leakage.

This procedure is applicable to horizontal multi-stage pumps, when bolted design for barrel cover is adopted.

This procedure is applicable to Nova Milanese and La Spezia plants.

2. Abbreviations

BB5: Between bearing pumps with a barrel casing as per API 610 standard.

3. General procedure: description

The bolted joint between the barrel casing and the discharge cover shall guarantee the packing of the pump's casings inside the barrel.

This is a critical part in the pump's design, involving the barrel casing and the discharge cover's sizing and the selection of the bolts' diameter and number.

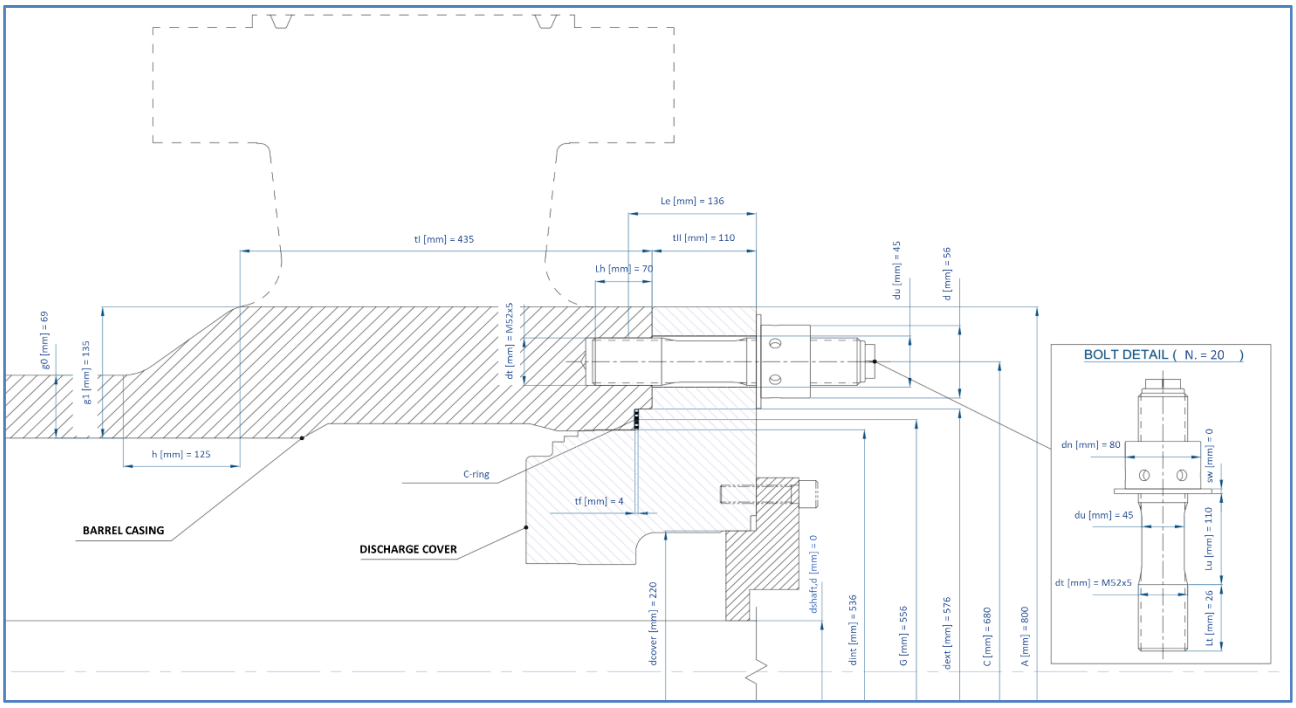

Figure 3.1: Typical bolted joint between barrel casing and discharge cover.

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4. Calculation method: bolted joint

The design and the verification of the bolted joint are carried out based on the values calculated as per the following chapters (refer to **SOP-477** and ***Mod.432***).

The calculation shall be carried out and verified for the following load cases:

1) Design fluid pressure at ambient temperature.
2) Design fluid pressure at design fluid temperature.
3) Hydrotest fluid pressure at ambient temperature.

4.1. Bolted joint description

The typical bolted joint geometry is shown in Figure 3.1.

Based on that geometry, the effective grip length of the threaded connection is defined as:

$$L_e = s_w + t + {}^{d_t}/_{2}$$

Where:

- L_e [mm] is the bolt effective total length.
- s_w [mm] is the washer thickness.
- t [mm] is the discharge cover thickness.
- d_t [mm] is the nominal diameter of the threaded part.

4.2. Fastener stiffness

The stiffness of the portion of a bolt or screw within the clamped zone generally consists of two parts: one of the unthreaded shank portion and one of the threaded portion. Hence, the overall stiffness of the bolt is equivalent to the stiffnesses of two springs in series, as per the following formula:

$$k_b = \frac{E_b A_t A_u}{A_t L_u + A_u L_t}$$

Where:

- k_b [N/mm] is the axial stiffness of the bolt.
- E_b [GPa] is the Young's modulus of the bolt's material.
- A_t [mm^2] is the tensile area of the threaded part of the bolt.
- A_u [mm^2] is the tensile area of the unthreaded part of the bolt.
- L_u [mm] is the length of the unthreaded part of the bolt.
- L_t [mm] is the length of the threaded part of the bolt.

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4.3. Barrel and cover stiffness

The equivalent stiffness of the barrel casing and the discharge cover in the clamped zone is based on the use of a simplified Rotscher's pressure-cone method with a fixed cone angle. If the clamped materials have identical Young's modulus, the overall stiffness is given by the following formula:

$$k_f = \frac{\pi\, E_f\, d\, tan\, \alpha}{2 \ln \frac{(l\, tan\, \alpha + d_n - d)(d_w + d)}{(l\, tan\, \alpha + d_n + d)(d_w - d)}}$$

Where:

- k_f [N/mm] is the barrel and cover's stiffness.
- E_f [GPa] is the Young's modulus of the barrel and cover's material.
- d [mm] is the diameter of the through hole in the discharge cover.
- α [°] is the half apex angle of the Rotscher's pressure cone.
- l [mm] is the overall thickness of the flanges, assumed equal to L_e.
- d_n [mm] is the nut's diameter.
- d_w [mm] is the diameter of the contact area between the bolt and the flange, assumed equal to d_n.

4.4. Tightening load calculation

The tightening force is proportional to the bolt yield stress, the tensile area and the tightening ratio:

$$F_0 = \xi \min(A_t, A_u)\ \sigma_{yb}$$

Where:

- F_0 [N] is the tightening force acting on each bolt of the flanged joint.
- ξ [%] is the tightening ratio, usually assumed as 70 - 80%.
- σ_{yb} [MPa] is the yield stress of the bolt material.

The effects of this force on the bolts and the flanges are calculated as follows:

$$\delta_b = \frac{F_0}{k_b}$$

$$\delta_f = \frac{F_0}{k_f}$$

$$\delta_0 = \delta_b + \delta_f$$

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Where:

- δ_b [mm] is the stud elongation due to the tightening force.
- δ_f [mm] is the flanges compression due to the tightening force.
- δ_0 [mm] is the overall elongation of the joint.

4.5. Thermal load calculation

In case of temperature variation, the bolts and the flanges expand.

If the thermal expansion coefficients of the bolts and the flanges' materials are the same, no thermal loads are expected.

On the contrary, if the thermal expansion coefficients are different, the temperature variation leads to a different joint loading. The change of the initial joint loading is calculated as:

$$F_t = \frac{k_b k_f}{k_b + k_f} \Delta T \, L_e \, (\alpha_b - \alpha_f)$$

Where:

- F_t [N] is the thermal force due to the temperature variation.
- ΔT [°C] is the temperature variation.
- α_b [1/°C] is the thermal expansion coefficient of the bolt's material.
- α_f [1/°C] is the thermal expansion coefficient of the barrel and cover's material.

In this case, the initial pre-load of the joint is calculated as the sum of the tightening and the thermal forces.

4.6. Internal pressure load calculation

When the fluid pressure is applied in the pump's casing, the following external force acts in each bolt of the flanged joint:

$$F_e = \frac{p}{N_{bolts}} \frac{\pi}{4} D^2$$

Where:

- F_e [N] is the resultant external force on each bolt.
- N_{bolts} is the number of bolts in the flanged joint.
- D [m] is the diameter of the discharge cover's wet surface, where the internal fluid pressure is applied.

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- p [barg] is the internal pressure. The following different load cases shall be considered in the design: the project design pressure and the hydrostatic pressure. The equivalent internal pressure shall be used, and it is calculated conservatively based on the following formula:

$$p_{eq} = p \cdot \frac{A_{tot}}{A_{cover}}$$

$$A_{tot} = \frac{\pi}{4} \cdot \left(D^2 - D_{shaft}^2\right)$$

$$A_{cover} = \frac{\pi}{4} \cdot \left(D^2 - D_{cover}^2\right)$$

Where:

- D_{cover} [m] is the cover internal diameter.
- D_{shaft} [m] is the shaft diameter. If no shaft is present, it is equal to 0.

In this case, the loads and the stresses of the bolts and the flanges are modified. The external load is distributed to the bolt and to the flange proportionally to their stiffnesses previously calculated:

$$F_e = F_b + F_f$$

$$\delta = \frac{F_b}{k_b} = \frac{F_f}{k_f} = \frac{F_e}{k_b + k_f}$$

$$F_b = \frac{k_b F_e}{k_b + k_f}$$

$$F_f = \frac{k_f F_e}{k_b + k_f}$$

Where:

- F_b [N] is the part of the external force that increases traction on the stud.
- F_f [N] is the part of the external force that reduces the flange contact pressure.

F_b increases the axial bolt stress, while F_f reduces the joint contact pressure. The bolt stresses are calculated as:

$$\sigma_{b0} = \frac{F_0}{A_r}$$

$$\sigma_b = \frac{F_0 + F_b}{A_r}$$

Where:

- σ_{b0} [MPa] is the axial bolt stress due to tightening and thermal load (if any).

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- σ_b [MPa] is the axial bolt stress due to tightening.
- A_r [mm^2] is the resistance area of the bolt, assumed as the minimum nonzero area between A_t and A_u.

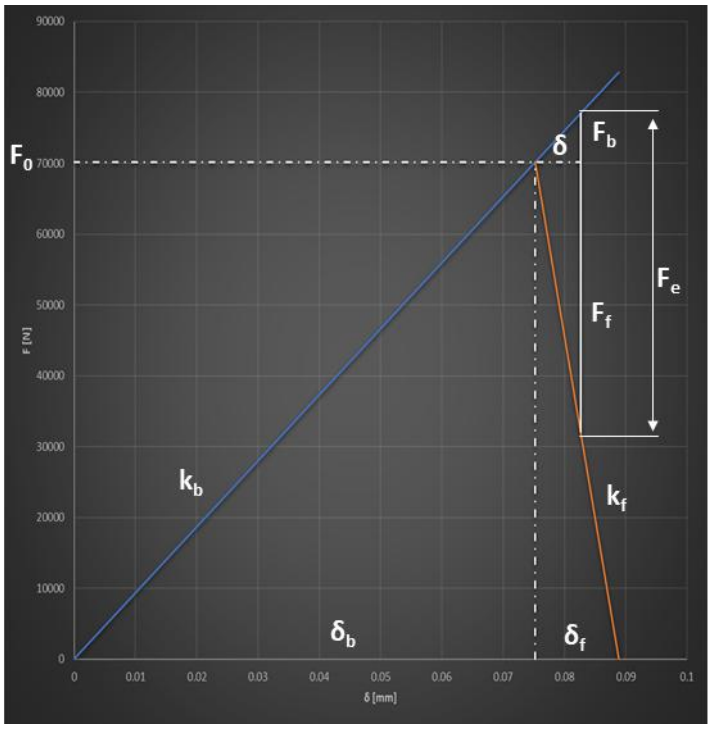

Figure 4.1: Force-deformation diagram with symbols and names of forces.

4.7. Lead screw thickness verification

The maximum allowable force on the barrel casing's thread is calculated as:

$$F_{max} = 75\% \, \pi \, d_t \, n_t \, p \frac{\sqrt{3}}{3} \sigma_{yf}$$

Where:

- F_{max} [N] is the maximum allowable load on the barrel casing's thread.
- n_t is the number of the gripping threads working in the barrel casing's threaded hole.
- p [mm] is the pitch of the thread.
- σ_{yf} [MPa] is the barrel and cover's material yield stress.

4.8. Gasket load

Even if the design is based on metal-to-metal contact, it is typical to install spiral wound gaskets, O-rings or C-rings between the barrel and the discharge cover.

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When an O-ring or C-ring is used, the force applied between the pump barrel and the cover is not significant compared to the pressure or tightening loads, therefore it is usually neglected.

The same consideration is not applicable for spiral wound gaskets. In this case, the gasket load acts on the cover in the same direction of the fluid pressure load.

The gasket load is calculated based on the loading and unloading curves, which are typical for each gasket type (reference plot is reported in *Figure 4.2*). The applicable load is determined based on the compression of the gasket during tightening, comparing the nominal and the assembled gasket thicknesses, and the subsequent expansion due to the cover deflection, calculated as per Section 5.1.

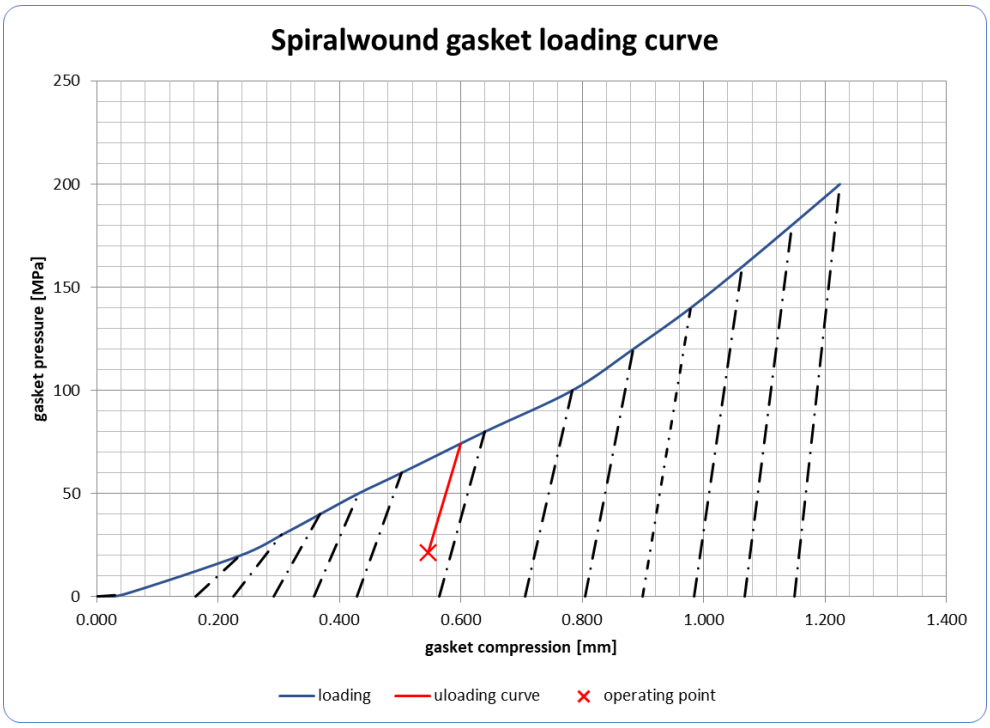

Figure 4.2: Reference loading and unloading curves for spiral wound gasket.

The data to create this diagram can be collected from gasket databases, such as The gasket database, or directly from the gasket manufacturer.

The information required is the following:

- The surface pressure at assembly condition Q_A [MPa].
- The unloading secant modulus of elasticity of the gasket E_G [MPa].
- The gasket thickness at assembly condition [mm].

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5. Calculation method: cover design

5.1. General approach

In order to evaluate the cover deflection under the applicable load conditions, the following approach is followed.

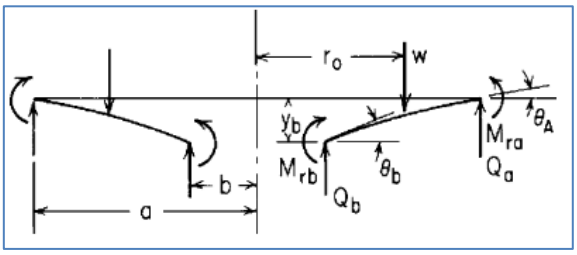

Figure 5.1: Reference drawing of annular plate with a uniform annular line load w at a radius r_0.

The pump cover is considered as a circular plate of uniform thickness (*Figure 5.1)* and the following formulas from section 11 of Roark's Formula for Stress and Strain are used.

$$y = y_b + \theta_b \cdot r \cdot F_1 + M_{rb} \cdot \frac{r^2}{D} \cdot F_2 + Q_b \cdot \frac{r^3}{D} \cdot F_3 - w \cdot \frac{r^3}{D} \cdot G_3$$

$$\theta = \theta_b \cdot F_4 + M_{rb} \cdot \frac{r}{D} \cdot F_5 + Q_b \cdot \frac{r^2}{D} \cdot F_6 - w \cdot \frac{r^2}{D} \cdot G_6$$

In addition, the following cases from Table 11.2 of Roark's Formula for Stress and Strain are considered:

- Case 1e: outer edge fixed, inner edge free with unit line load.

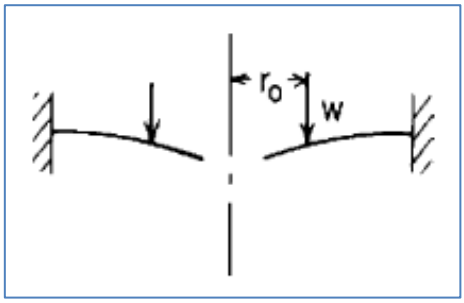

Figure 5.2: Case 1e from Table 11.2 of Roark's Formula for Stress and Strain.

$$y_a = 0$$

$$y_b = -\frac{w \cdot a^3}{D} \cdot \left(\frac{C_1 \cdot L_6}{C_4} - L_3\right)$$

$$\theta_a = 0$$

$$\theta_b = \frac{w \cdot a^2}{D \cdot C_4} \cdot L_6$$

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$$M_{ra} = -w \cdot a \cdot \left(L_9 - \frac{C_7 \cdot L_6}{C_4}\right)$$

$$M_{rb} = 0$$

- Case 2e: outer edge fixed, inner edge free with uniform pressure.

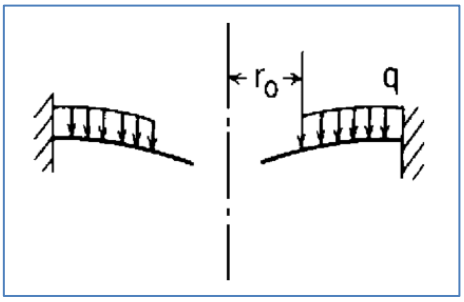

Figure 5.3 Case 2e from Table 11.2 of Roark's Formula for Stress and Strain.

$$y_a = 0$$

$$y_b = -\frac{q \cdot a^4}{D} \cdot \left(\frac{C_1 \cdot L_{14}}{C_4} - L_{11}\right)$$

$$\theta_a = 0$$

$$\theta_b = \frac{q \cdot a^3}{D \cdot C_4} \cdot L_{14}$$

$$M_{ra} = -q \cdot a^2 \cdot \left(L_{17} - \frac{C_7 \cdot L_{14}}{C_4}\right)$$

$$M_{rb} = 0$$

The coefficients indicated in the previous formulas are reported below:

$$C_1 = \frac{1+\nu}{2} \cdot \frac{b}{a} \cdot \ln\frac{a}{b} + \frac{1-\nu}{2} \cdot \left(\frac{a}{b} - \frac{b}{a}\right)$$

$$C_4 = \frac{1}{2} \cdot \left[(1+\nu) \cdot \frac{b}{a} + (1-\nu) \cdot \frac{a}{b}\right]$$

$$C_7 = \frac{1}{2} \cdot (1-\nu^2) \cdot \left(\frac{a}{b} - \frac{b}{a}\right)$$

$$D = \frac{E \cdot t^3}{12} \cdot (1-\nu^2)$$

$$F_1 = \frac{1+\nu}{2} \cdot \frac{b}{r} \cdot \ln\frac{r}{b} + \frac{1-\nu}{2} \cdot \left(\frac{r}{b} - \frac{b}{r}\right)$$

$$F_2 = \frac{1}{4} \cdot \left[1 - \left(\frac{b}{r}\right)^2 \cdot \left(1 + 2\ln\left(\frac{r}{b}\right)\right)\right]$$

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$$F_3 = \frac{b}{4r} \cdot \left\{ \left[\left(\frac{b}{r}\right)^2 + 1 \right] \cdot \ln\frac{r}{b} + \left(\frac{b}{r}\right)^2 - 1 \right\}$$

$$F_4 = \frac{1}{2} \cdot \left[(1+\nu) \cdot \frac{b}{r} + (1-\nu) \cdot \frac{r}{b} \right]$$

$$F_5 = \frac{1}{2} \cdot \left[1 - \left(\frac{b}{r}\right)^2 \right]$$

$$F_6 = \frac{b}{4r} \cdot \left[\left(\frac{b}{r}\right)^2 - 1 + 2\ln\frac{r}{b} \right]$$

$$G_3 = \frac{r_0}{4r} \cdot \left\{ \left[\left(\frac{r_0}{r}\right)^2 + 1 \right] \cdot \ln\frac{r}{r_0} + \left(\frac{r_0}{r}\right)^2 - 1 \right\} \cdot \langle r - r_0 \rangle^0$$

$$G_6 = \frac{r_0}{4r} \cdot \left[\left(\frac{r_0}{r}\right)^2 - 1 + 2\ln\frac{r}{r_0} \right] \cdot \langle r - r_0 \rangle^0$$

$$G_9 = \frac{r_0}{r} \cdot \left\{ \frac{1+\nu}{2} \cdot \ln\frac{r}{r_0} + \frac{1-\nu}{4} \cdot \left[1 - \left(\frac{r_0}{r}\right)^2 \right] \right\} \cdot \langle r - r_0 \rangle^0$$

$$G_{11} = \frac{1}{64} \cdot \left\{ 1 + 4 \cdot \left(\frac{r_0}{r}\right)^2 - 5 \cdot \left(\frac{r_0}{r}\right)^4 - 4 \cdot \left(\frac{r_0}{r}\right)^2 \cdot \left[2 + \left(\frac{r_0}{r}\right)^2 \right] \cdot \ln\frac{r}{r_0} \right\} \cdot \langle r - r_0 \rangle^0$$

$$G_{14} = \frac{1}{16} \cdot \left[1 - \left(\frac{r_0}{r}\right)^4 - 4 \cdot \left(\frac{r_0}{r}\right)^2 \cdot \ln\frac{r}{r_0} \right] \cdot \langle r - r_0 \rangle^0$$

$$G_{17} = \frac{1}{4} \cdot \left\{ 1 - \frac{1-\nu}{4} \cdot \left[1 - \left(\frac{r_0}{r}\right)^4 \right] - \left(\frac{r_0}{r}\right)^2 \cdot \left[1 + (1+\nu) \cdot \ln\frac{r}{r_0} \right] \right\} \cdot \langle r - r_0 \rangle^0$$

$$L_3 = \frac{r_0}{4a} \cdot \left\{ \left[\left(\frac{r_0}{a}\right)^2 + 1 \right] \cdot \ln\frac{a}{r_0} + \left(\frac{r_0}{a}\right)^2 - 1 \right\}$$

$$L_6 = \frac{r_0}{4a} \cdot \left[\left(\frac{r_0}{a}\right)^2 - 1 + 2\ln\frac{a}{r_0} \right]$$

$$L_9 = \frac{r_0}{a} \cdot \left\{ \frac{1+\nu}{2} \cdot \ln\frac{a}{r_0} + \frac{1-\nu}{4} \cdot \left[1 - \left(\frac{r_0}{a}\right)^2 \right] \right\}$$

$$L_{11} = \frac{1}{64} \cdot \left\{ 1 + 4 \cdot \left(\frac{r_0}{a}\right)^2 - 5 \cdot \left(\frac{r_0}{a}\right)^4 - 4 \cdot \left(\frac{r_0}{a}\right)^2 \cdot \left[2 + \left(\frac{r_0}{a}\right)^2 \right] \cdot \ln\frac{a}{r_0} \right\}$$

$$L_{14} = \frac{1}{16} \cdot \left[1 - \left(\frac{r_0}{a}\right)^4 - 4 \cdot \left(\frac{r_0}{a}\right)^2 \cdot \ln\frac{a}{r_0} \right]$$

$$L_{17} = \frac{1}{4} \cdot \left\{ 1 - \frac{1-\nu}{4} \cdot \left[1 - \left(\frac{r_0}{a}\right)^4 \right] - \left(\frac{r_0}{a}\right)^2 \cdot \left[1 + (1+\nu) \cdot \ln\frac{a}{r_0} \right] \right\}$$

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Where:

- a [mm] is the outer radius of the plate.
- b [mm] is the inner radius of the plate.
- E [MPa] is the Young's modulus of the plate material.
- r [mm] is the reference radius of the plate.
- M_{ra} [N] and M_{rb} [N] are the unit radial bending moments applied at the outer radius a and at the inner radius b.
- q [MPa] is the load per unit area.
- Q_b [N/mm] is the unit shear force (force per unit of circumferential length) applied at the inner radius b.
- r_0 [mm] is the radial location of unit line loading.
- t [mm] is the thickness of the plate.
- y [mm] is the displacement at the reference radius r [mm] of the plate.
- y_a [mm] and y_b [mm] are the flange displacements at the outer radius a and at the inner radius b.
- w [N/mm] is the unit line load (force per unit of circumferential length) applied at radius r_0.
- θ [rad] is the radial slope of the plate.
- θ_a [rad] and θ_b [rad] are the radial slopes of the plate at the outer radius a and at the inner radius b.
- v [-] is the Poisson's ratio.
- $\langle r - r_0 \rangle^0$ is a term equal to 1 if $r > r_0$ and to 0 in the other cases.

The formulas reported above are used to estimate the deflection of the pump cover by using the superposition method. The applicable loads are the bolting (thermal and tightening) and the gasket reaction forces calculated as per Case 1e and the internal pressure calculated as per Case 2e.

Furthermore, the following assumption is made: the radius a is defined as that of the innermost circumference where the cover and barrel are in metal-to-metal contact. It is determined by applying the conservative boundary condition, that the unit radial bending moment (Mra) is equal to zero.

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5.2. Approach as per ASME-BPVC Sect. VIII Div. 1

The pump cover is considered equivalent to a flat circular head without opening. For this reason, the calculation method adopted for the preliminary design is based on Appendix Y of ASME-BPVC Sect. VIII Div. 1. This appendix applies to the flat face flanges with metal-to-metal contact outside the bolt circle. The applicable equations are those relevant to the following flange class and category:

- Class 3 Assembly: Any assemblage consisting of a reducer or a flat circular head without an opening or with a central, reinforced opening provided the diameter of the opening in the reducing flange or flat cover is less than one-half of the bolt circle diameter. In the analysis, the reducing flange is the equivalent of a flat circular head without an opening. *Figure 5.4* illustrates configuration of a Class 3 assembly.
- Category 1 Flange: an integral flange or an optional flange calculated as an integral flange.

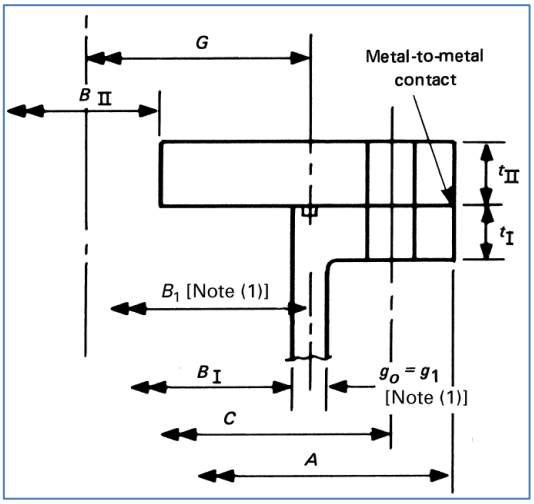

Figure 5.4: Class 3 Assembly - Category 1 Flange.

As per the summary table Y-6.1 of ASME-BVPC Sect. VIII Div. 1, the following applicable equations are used for Class 3 Assembly and Category 1 Flange:

$$C_1 = -\frac{0.748-1.567\cdot J_S\cdot log\left(A/B_1\right)}{1+1.3\cdot J_S} \text{ (Y-3 eq. 1)}$$

$$C_2 = \frac{\pi/32\cdot P\cdot B_1-1.3\cdot J_P\cdot M_P}{1+1.3\cdot J_S} \quad \text{(Y-3 eq. 2)}$$

$$C_3 = -\frac{0.575-1.206\cdot J_S\cdot log\left(A/B_1\right)}{1+1.3\cdot J_S} \text{ (Y-3 eq. 3)}$$

$$C_4 = -\frac{J_P\cdot M_P}{J_S+t_I^3/F_I'} \text{ (Y-3 eq. 4)}$$

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$$J_S = \frac{1}{B_1} \cdot \left[\frac{2h_D}{\beta} + \frac{h_C}{a}\right] + \pi \cdot r_B$$

$$J_P = \frac{1}{B_1} \cdot \left[\frac{h_D}{\beta} + \frac{h_C}{a}\right] + \pi \cdot r_B$$

$$F_I' = g_0^2 \frac{h_0 + Ft}{V} \text{(Y-3 eq. 6a)}$$

Rigid body rotation of flanges times E*:

$$E_I^* \theta_{rbI} = \frac{X \cdot (C_4 - C_2)}{1.206 \cdot \log\left(A/B_1\right) - X \cdot C_3 - (1-X) \cdot C_1} \text{ (Y-6.3 eq. 17)}$$

$$E_{II}^* \theta_{rbII} = -E_I^* \theta_{rbI} \cdot \left(E_{II}^* / E_I^*\right) \quad \text{(Y-6.3 eq. 18)}$$

Total flange moment at diameter B_1:

$$M_{S\,I} = C_3 \cdot (E_I^* \theta_{rbI}) + C_4 \quad \text{(Y-6.3 eq. 19)}$$

$$M_{S\,II} = C_1 \cdot (E_{II}^* \theta_{rbII}) + C_2 \quad \text{(Y-6.3 eq. 20)}$$

Unbalanced flange moment at diameter B_1:

$$M_{u\,I} = 1.206 \cdot E_I^* \theta_{rbI} \cdot \log\left(A/B_1\right) \text{ (Y-6.3 eq. 21)}$$

$$M_{u\,II} = 1.206 \cdot E_{II}^* \theta_{rbII} \cdot \log\left(A/B_1\right) \text{ (Y-6.3 eq. 22)}$$

Balanced flange moment at diameter B_1:

$$M_{b\,I} = M_{S\,I} - M_{u\,I} \text{ (Y-6.3 eq. 23)}$$

$$M_{b\,II} = M_{S\,II} - M_{u\,II} \text{ (Y-6.3 eq. 24)}$$

Slope of flange at diameter B_1 times E:

$$E_I \theta_{B\,I} = \frac{5.46}{\pi t_I^3} \cdot (J_S M_{b\,I} + J_P M_P) + E_I^* \theta_{rbI} / t_I^3 \quad \text{(Y-6.3 eq. 25)}$$

$$E_{II} \theta_{B\,II} = \frac{-1.337(M_{S\,II} - \pi P B_1^3/32)}{t_{II}^3} \text{ (Y-6.3 eq. 26)}$$

Contact force between flanges at h_C:

$$H_C = (M_P + M_{b\,I})/h_C \text{ (Y-6.3 eq. 27)}$$

Bolt lead at operating condition:

$$W_{m1} = H + H_G + H_C \text{ (Y-6.3 eq. 28)}$$

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Operating bolt stress:

$$\sigma_b = W_{m1}/A_b \quad \text{(Y-6.3 eq. 29)}$$

Design prestress in bolts:

$$S_i = \sigma_b - \frac{1.159 h_C^2 (M_P + M_{b\,I})}{2(1-X) a t_I^3 l r_{E\,I} B_1} \quad \text{(Y-6.3 eq. 30)}$$

Radial stress in flange I at bolt circle:

$$S_{R\,I} = \frac{6(M_P + M_{S\,I})}{t_I^2 (\pi C - nD)} \quad \text{(Y-6.3 eq. 31)}$$

Radial stress in flange I at inside diameter:

$$S_{R\,I} = -\left(\frac{2Ft_I}{h_0 + Ft_I} + 6\right) \cdot \frac{M_{S\,I}}{\pi B_1 t_I^2} \quad \text{(Y-6.3 eq. 32a)}$$

Tangential stress in flange I at inside diameter:

$$S_{T\,I} = \frac{t_I E_I \theta_{B\,I}}{B_1} + \left(\frac{2Ft_I Z}{h_0 + Ft_I} - 1.8\right) \frac{M_{S\,I}}{\pi B_1 t_I^2} \quad \text{(Y-6.3 eq. 33a)}$$

Longitudinal hub stress in flange I:

$$S_{H\,I} = \frac{h_0 E_I \theta_{B\,I} f}{0.91 \left(\frac{g_1}{g_0}\right)^2 B_1 V} \quad \text{(Y-6.3 eq. 34a)}$$

Radial stress in flange II at bolt circle:

$$S_{R\,II} = \frac{6(M_P + M_{S\,II})}{t_{II}^2 (\pi C - nD)} \quad \text{(Y-6.3 eq. 35)}$$

Radial stress in flange II at diameter B_1:

$$S_{R\,II} = \frac{6 M_{S\,II}}{\pi B_1 t_{II}^2} \quad \text{(Y-6.3 eq. 36)}$$

Tangential stress in flange II at diameter B_1:

$$S_{T\,II} = \frac{t_{II} E_{II} \theta_{B\,II}}{B_1} - \frac{1.8 M_{S\,II}}{\pi B_1 t_{II}^2} \quad \text{(Y-6.3 eq. 37)}$$

Tangential stress in flange II at center of flange II:

$$S_{R\,II} = S_{T\,II} = \frac{0.3094 P B_1^2}{t_{II}^2} - \frac{1.8 M_{S\,II}}{\pi B_1 t_{II}^2} \quad \text{(Y-6.3 eq. 38)}$$

Where:

- A [mm] is the outside diameter of the flange.
- a is the shape factor, calculated as $(A + C)/2B1$.

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- AR is the bolt hole aspect ratio, calculated as $nD/\pi C$.
- B [mm] is the inside diameter of the flange.
- B_1 [mm] is the equivalent inside diameter of the flange. It is calculated as:
  - $B + g1$: for integral type flanges that have calculated values h/h_0 and g_1/g_0 which would indicate f≤1, although the minimum permitted value is 1.
  - $B + g0$: for integral type flanges when $f \geq 1$.
- C [mm] is the bolt circle diameter.
- D [mm] is the diameter of the bolt hole.
- d is the factor equal to $\frac{U}{V}\ h_0\ g_0^2$ for the integral type flanges.
- E [MPa] is the modulus of elasticity of flange material, corrected for operating temperature.
- E_I^* [Nmm] is the factor $E_I t_I^3$.
- E_{II}^* [Nmm] is the factor $E_{II} t_{II}^3$.
- e is the factor F/h_0 for the integral type flanges.
- F is the factor for integral type flanges as per Appendix 2 of ASME-BVPC Sect. VIII Div. 1.
- f is the hub stress correction factor for integral flanges from Appendix 2 Figure 2-7.6 of ASME-BVPC Sect. VIII Div. 1.
- G [mm] is the diameter at location of gasket load reaction, equal to the mean diameter of the gasket.
- g_0 [mm] is the thickness of hub at small end.
- g_1 [mm] is the thickness of hub at back of flange.
- H [N] is the total hydrostatic end force, equal to $0.785\ G^2 P$.
- H_D [N] is the hydrostatic end force on area inside of flange, equal to $0.785\ B^2 P$.
- H_G [N] is the gasket load due to seating pressure, plus axial force generated by self-sealing of gasket.
- H_T [N] is the difference between total hydrostatic end force and the hydrostatic end force on area inside of flange, equal to $H - H_D$.
- h [mm] is the hub length.
- h_C [mm] is the radial distance from bolt circle to flange-spacer or flange-flange bearing circle where tangential contact occurs. Tangential contact exists from the selected value of h_C to h_{Cmax}.
- h_D [mm] is the radial distance from the bolt circle, to the circle on which H_D acts.
- h_G [mm] is the radial distance from gasket load reaction to the bolt circle, equal to $(C - G)/2$.
- h_0 [mm] is the factor equal to $\sqrt{B\ g_0}$.

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- h_T [mm] is the radial distance from the bolt circle to the circle on which H_T acts.
- K is the ratio of outside diameter of flange to inside diameter of flange, equal to A/B.
- l is the calculated strain length of the bolt, equal to $t_I + t_{II} + (0.5d_b for\ each\ theaded\ end)$.
- M_P [Nm] is the moment due to H_D, H_T, H_G, equal to $M_D + M_T + M_G = H_D h_D + H_T h_T + H_G h_G$.
- P [MPa] is the internal design pressure.
- r_E is the elasticity factor, calculated as modulus of elasticity of flange material divided by modulus of elasticity of bolting material, corrected for operating temperature.
- r_B is the bolt – hole flexibility factor, calculated as:

$$r_B = \frac{1}{n} \cdot \left[\frac{4}{\sqrt{1-\overline{AR}^2}} \tan^{-1} \sqrt{\frac{1+\overline{AR}}{1-\overline{AR}}} - \pi - 2\overline{AR} \right]$$

- t_I and t_{II} [mm] are the thicknesses respectively of the non-reducing flange and of the reducer or flat circular head in a Class 3 assembly.
- V is the factor for integral type flanges as per Appendix 2 of ASME-BVPC Sect. VIII Div. 1.
- β is the shape factor for full face metal-to-metal contact flanges, calculated as $(C + B1)/2B1$.

The f, F and V coefficients are extracted from *Figure 5.5*, *Figure 5.6* and *Figure 5.7* as per Appendix 2 of ASME-BVPC Sect. VIII Div. 1.

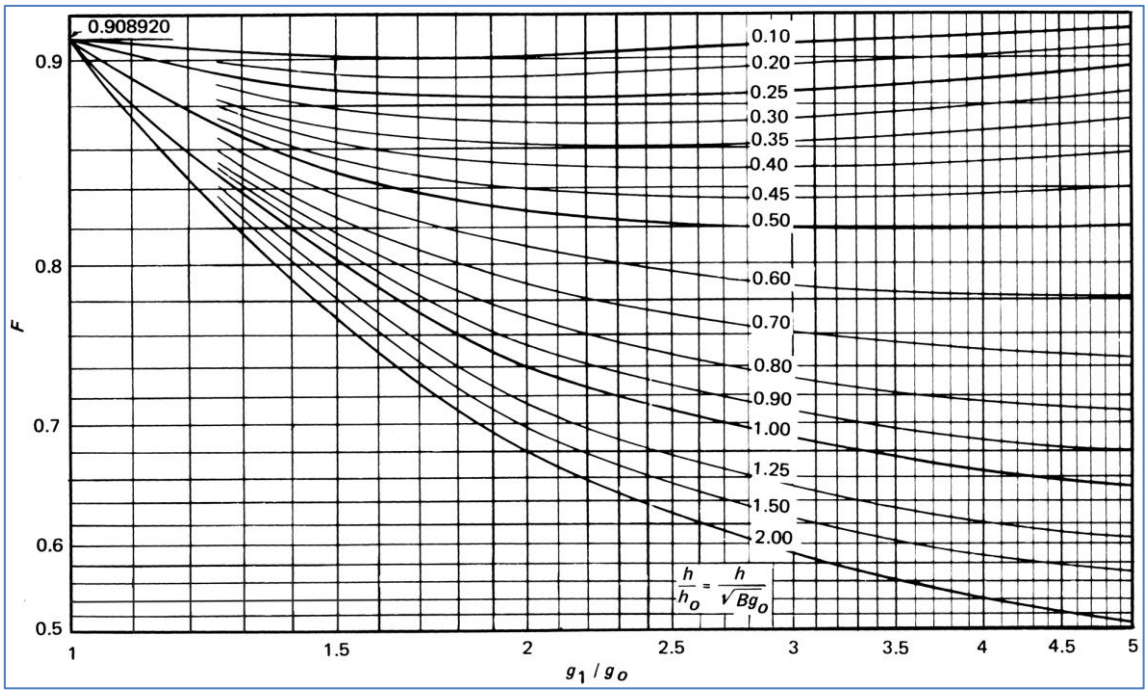

Figure 5.5: Values of F (integral flange factors) - Appendix 2 Figure 2-7.2 of ASME-BVPC Sect. VIII Div. 1.

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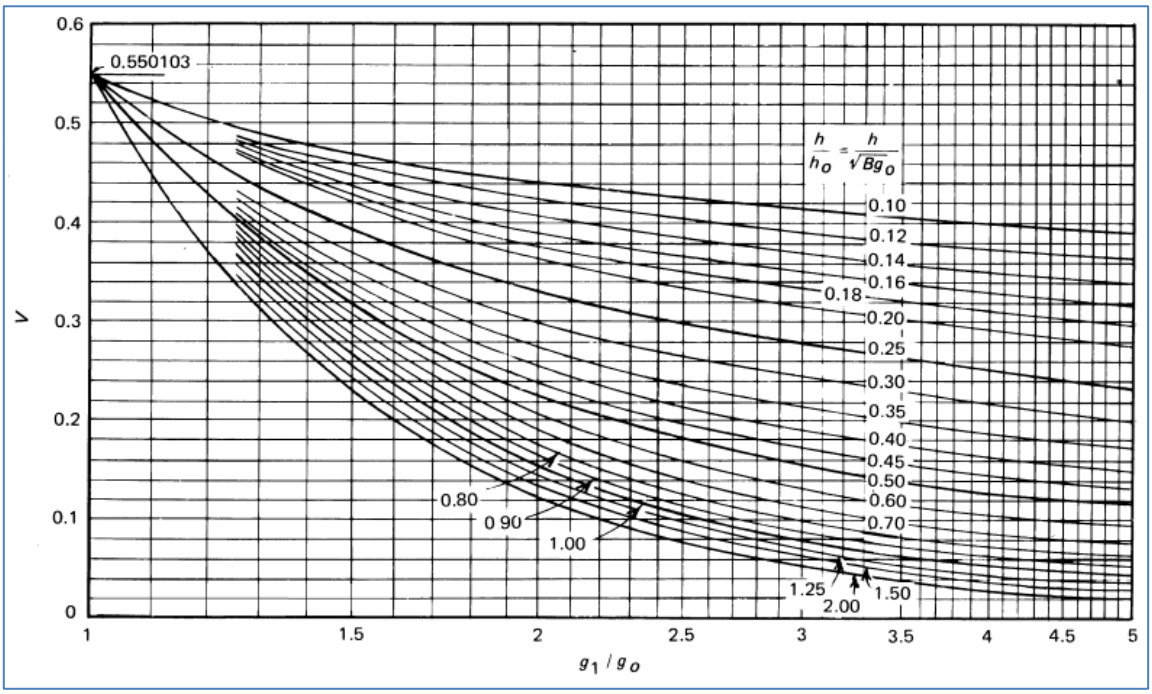

Figure 5.6: Values of V (integral flange factors) - Appendix 2 Figure 2-7.3 of ASME-BVPC Sect. VIII Div. 1.

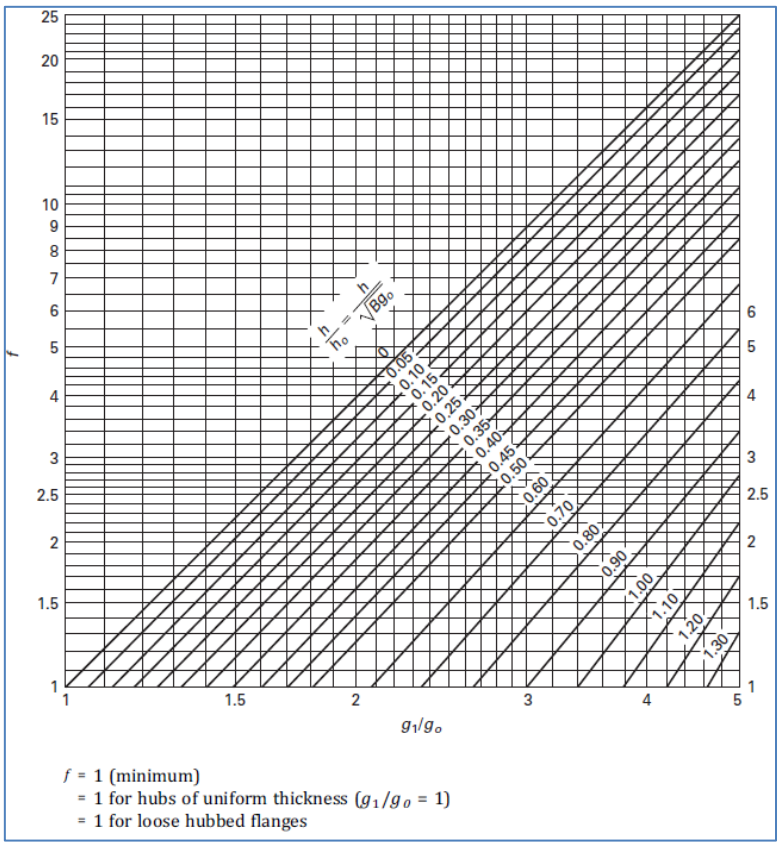

Figure 5.7: Values of f (hub stress correction factor) - Appendix 2 Figure 2-7.6 of ASME-BVPC Sect. VIII Div. 1.

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Such coefficients can be calculated by using the equations from Table 2-7.1 of ASME-BVPC Sect. VIII Div. 1:

$$F = -\frac{E_6}{\left(\frac{C}{2.73}\right)^{\frac{1}{4}} \frac{(1+A)^3}{C}}$$

$$V = \frac{E_4}{\left(\frac{2.73}{C}\right)^{\frac{1}{4}} (1+A)^3}$$

$$f = \frac{C_{36}}{1+A}$$

The above-mentioned coefficients are calculated based on the following formulas:

$$(1)\ A = \frac{g_1}{g_0} - 1$$

$$(2)\ C = 43.68\left(\frac{h}{h_0}\right)^4$$

$$(3)\ C_1 = \frac{1}{3} + \frac{A}{12}$$

$$(4)\ C_2 = \frac{5}{42} + \frac{17A}{336}$$

$$(5)\ C_3 = \frac{1}{210} + \frac{A}{360}$$

$$(6)\ C_4 = \frac{11}{360} + \frac{59A}{5040} + \frac{1 + 3A}{C}$$

$$(7)\ C_5 = \frac{1}{90} + \frac{5A}{1008} - \frac{(1 + A)^3}{C}$$

$$(8)\ C_6 = \frac{1}{120} + \frac{17A}{5040} + \frac{1}{C}$$

$$(9)\ C_7 = \frac{215}{2772} + \frac{51A}{1232} + \frac{\frac{60}{7} + \frac{225A}{14} + \frac{75A^2}{7} + \frac{5A^3}{2}}{C}$$

$$(10)\ C_8 = \frac{31}{6930} + \frac{128A}{45{,}045} + \frac{\frac{6}{7} + \frac{15A}{7} + \frac{12A^2}{7} + \frac{5A^3}{11}}{C}$$

$$(11)\ C_9 = \frac{533}{30{,}240} + \frac{653A}{73{,}920} + \frac{\frac{1}{2} + \frac{33A}{14} + \frac{39A^2}{28} + \frac{25A^3}{84}}{C}$$

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$$(12)\ C_{10} = \frac{29}{3780} + \frac{3A}{704} - \frac{\frac{1}{2} + \frac{33A}{14} + \frac{81A^2}{28} + \frac{13A^3}{12}}{C}$$

$$(13)\ C_{11} = \frac{31}{6048} + \frac{1763A}{665{,}280} + \frac{\frac{1}{2} + \frac{6A}{7} + \frac{15A^2}{28} + \frac{5A^3}{42}}{C}$$

$$(14)\ C_{12} = \frac{1}{2925} + \frac{71A}{300{,}300} + \frac{\frac{8}{35} + \frac{18A}{35} + \frac{156A^2}{385} + \frac{6A^3}{55}}{C}$$

$$(15)\ C_{13} = \frac{761}{831{,}600} + \frac{937A}{1{,}663{,}200} + \frac{\frac{1}{35} + \frac{6A}{35} + \frac{11A^2}{70} + \frac{3A^3}{70}}{C}$$

$$(16)\ C_{14} = \frac{197}{415{,}800} + \frac{103A}{332{,}640} - \frac{\frac{1}{35} + \frac{6A}{35} + \frac{17A^2}{70} + \frac{A^3}{10}}{C}$$

$$(17)\ C_{15} = \frac{233}{831{,}600} + \frac{97A}{554{,}400} + \frac{\frac{1}{35} + \frac{3A}{35} + \frac{A^2}{14} + \frac{2A^3}{105}}{C}$$

$$(18)\ C_{16} = C_1C_7C_{12} + C_2C_8C_3 + C_3C_8C_2 - (C_3^2C_7 + C_8^2C_1 + C_2^2C_{12})$$

$$(19)\ C_{17} = \frac{C_4C_7C_{12} + C_2C_8C_{13} + C_3C_8C_9 - (C_{13}C_7C_3 + C_8^2C_4 + C_{12}C_2C_9)}{C_{16}}$$

$$(20)\ C_{18} = \frac{[C_5C_7C_{12} + C_2C_8C_{14} + C_3C_8C_{10} - (C_{14}C_7C_3 + C_8^2C_5 + C_{12}C_2C_{10})]}{C_{16}}$$

$$(21)\ C_{19} = \frac{[C_6C_7C_{12} + C_2C_8C_{15} + C_3C_8C_{11} - (C_{15}C_7C_3 + C_8^2C_6 + C_{12}C_2C_{11})]}{C_{16}}$$

$$(22)\ C_{20} = \frac{[C_1C_9C_{12} + C_4C_8C_3 + C_3C_{13}C_2 - (C_3^2C_9 + C_{13}C_8C_1 + C_{12}C_4C_2)]}{C_{16}}$$

$$(23) C_{21} = [C_1C_{10}C_{12} + C_5C_8C_3 + C_3C_{14}C_2 - (C_3^2C_{10} + C_{14}C_8C_1 + C_{12}C_{15}C_2)]/C_{16}$$

$$(24)\ C_{22} = \frac{[C_1C_{11}C_{12} + C_6C_8C_3 + C_3C_{15}C_2 - (C_3^2C_{11} + C_{15}C_8C_1 + C_{12}C_6C_2)]}{C_{16}}$$

$$(25)\ C_{23} = \frac{[C_1C_7C_{13} + C_2C_9C_3 + C_4C_8C_2 - (C_3C_7C_4 + C_8C_9C_1 + C_{22}C_{13})]}{C_{16}}$$

$$(26) C_{24} = \frac{[C_1C_7C_{14} + C_2C_{10}C_3 + C_5C_8C_2 - (C_3C_7C_5 + C_8C_{10}C_1 + C_{22}C_{14})]}{C_{16}}$$

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$$(27)\ C_{25} = \frac{[C_1C_7C_{15} + C_2C_{11}C_3 + C_6C_8C_2 - (C_3C_7C_6 + C_8C_{11}C_1 + C_{22}C_{15})]}{C_{16}}$$

$$(28)\ C_{26} = -\left(\frac{C}{4}\right)^{\frac{1}{4}}$$

$$(29)\ C_{27} = C_{20} - C_{17} - \frac{5}{12} + C_{17}C_{26}$$

$$(30)\ C_{28} = C_{22} - C_{19} - \frac{1}{12} + C_{19}C_{26}$$

$$(31)\ C_{29} = -\left(\frac{C}{4}\right)^{\frac{1}{2}}$$

$$(32)\ C_{30} = -\left(\frac{C}{4}\right)^{\frac{3}{4}}$$

$$(33)\ C_{31} = \frac{3A}{2} - C_{17}C_{30}$$

$$(34)\ C_{32} = \frac{1}{2} - C_{19}C_{30}$$

$$(35)\ C_{33} = 0.5C_{26}C_{32} + C_{28}C_{31}C_{29} - (0.5C_{30}C_{28} + C_{32}C_{27}C_{29})$$

$$(36)\ C_{34} = \frac{1}{12} + C_{18} - C_{21} - C_{18}C_{26}$$

$$(37)\ C_{35} = -C_{18}\left(\frac{C}{4}\right)^{\frac{3}{4}}$$

$$(38)\ C_{36} = \frac{C_{28}C_{35}C_{29} - C_{32}C_{34}C_{29}}{C_{33}}$$

$$(39)\ C_{37} = \frac{[0.5C_{26}C_{35} + C_{34}C_{31}C_{29} - (0.5C_{30}C_{34} + C_{35}C_{27}C_{29})]}{C_{33}}$$

$$(40)\ E_1 = C_{17}C_{36} + C_{18} + C_{19}C_{37}$$

$$(41)\ E_2 = C_{20}C_{36} + C_{21} + C_{22}C_{37}$$

$$(42)\ E_3 = C_{23}C_{36} + C_{24} + C_{25}C_{37}$$

$$(43)\ E_4 = \frac{1}{4} + \frac{C_{37}}{12} + \frac{C_{36}}{4} - \frac{E_3}{5} - \frac{3E_2}{2} - E_1$$

$$(44)\ E_5 = E_1\left(\frac{1}{2} + \frac{A}{6}\right) + E_2\left(\frac{1}{4} + \frac{11A}{84}\right) + E_3\left(\frac{1}{70} + \frac{A}{105}\right)$$

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$$(45)\ E_6 = E_5 - C_{36}\left(\frac{7}{120} + \frac{A}{36} + \frac{3A}{C}\right) - \frac{1}{40} - \frac{A}{72} - C_{37}\left(\frac{1}{60} + \frac{A}{120} + \frac{1}{C}\right)$$

In general, the above reported equations from (1) to (45) do not need to be solved when $g_1 = g_0$. In this case, the coefficients are assumed as $F = 0.908920$, $V = 0.550103$ and $f = 1$.

In addition to the above, the flange rigidity verification is carried out to check that the flanges are sufficiently rigid with no significant deformation, which may lead to leakage. This verification is based on Appendix 2 Table 2-14 of ASME-BVPC Sect. VIII Div. 1. The Flange Rigidity Factor for integral type flanges is calculated as:

$$J = \frac{52.14 V M_0}{L E g_0^2 K_I h_0}$$

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Where:

- E is the modulus of elasticity for the flange material at design temperature (operating condition) or at atmospheric temperature.
- K_I is the rigidity factor for integral or optional flange types, equal to 0.3.
- L is the factor equal to $\frac{te+1}{T} + \frac{t^3}{d}$.
- M_0 [Nm] is the total moment acting upon the flange.
- V is the factor for integral type flanges as per Appendix 2 of ASME-BVPC Sect. VIII Div. 1.

6. Calculation method: barrel thickness

The verification of the barrel's thickness is carried out based on section UG-27 as per ASME-BVPC Sect. VIII Div. 1. It provides the calculation method to estimate the minimum required thickness of shells under internal pressure.

The minimum acceptable thickness is obtained through the following formula:

$$t_{min} = \frac{PR}{SE - 0.6P}$$

Where:

- P [MPa] is the design pressure.
- R [mm] is the inside radius of the shell (barrel).
- S [MPa] is the maximum allowable stress.
- E is the joint efficiency for an appropriate joint in cylindrical shells. If the barrel is cast, E is equal to 1. Otherwise, it can be selected from Table UW-12 as per ASME-BVPC Sect. VIII Div. 1.

The formula mentioned above is reliable if the following criteria are met:

$$t \leq 1.5 \cdot R \quad \text{or} \quad P \leq 0.385 \cdot S \cdot E$$

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7. Acceptability criteria

The following criteria shall be verified in all the load cases.

7.1. Bolted joint verification

The acceptance criteria for the bolted joint verification are the following:

1. Verification of the axial stress:

$$\sigma_b \leq 90\% \, \sigma_{yb}$$

2. Verification of the reduction of contact pressure between the barrel casing and the discharge cover:

$$F_f \leq 60\% \, F_0$$

3. Verification of the barrel casing's threaded hole:

$$F_0 + F_b \leq 80\% \, F_{max}$$

7.2. Discharge cover verification

The discharge cover deflection shall meet the following criteria:

1. The deflection at the gasket diameter shall be below the maximum allowable limit to guarantee sealing and prevent any leakage.
2. Based on the position of the detachment circle (radius *a*) between the cover and the barrel calculated as per Section 5.1, the following conditions are identified:
   a. It is within the bolt circle diameter: the solution is acceptable.
   b. It is between the bolt circle and the cover external diameters: the solution is still acceptable, but more detailed analysis is recommended.
   c. It is outside the cover external diameter: the solution is not acceptable.

In addition, the acceptance criteria for the bolted joint's stress verification, based on Y-7 as per ASME-BVPC Sect. VIII Div. 1, are the following:

1. Longitudinal hub stress S_H must not be greater than 1.5 x S_f for materials other than cast iron. In case of Category 1 or 2 cast iron flanges, longitudinal hub stress S_H shall not be greater than S_f.
   Longitudinal hub stress S_H must not be greater than the smaller value between 1.5 x S_f or 2.5 x S_n for Category 1 flanges where the pipe or shell constitutes the hub.
   Longitudinal hub stress S_H must not be greater than the smaller value between 1.5 x S_f or 2.5 x S_n for integral Nonmandatory Appendix Y flanges (Category 1).
2. Radial stress S_R must not be greater than S_f.
3. Tangential stress S_T must not be greater than S_f.

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4. $(S_H + S_R)/2$ must not be greater than S_f and $(S_H + S_T)/2$ must not be greater than S_f.
5. S_R and S_T at the center of the reducing flange in a Class 3 assembly shall not exceed S_f.

The rigidity verification is carried out based on the Appendix 2, Table 2-14 as per ASME-BVPC Sect. VIII Div. 1, requiring that the Flange Rigidity Factor is not greater than 1 ($J \leq 1$).

As general recommendation, these criteria must be used during the pre-design. However, more detailed calculations through FEM analysis may be done.

7.3. Barrel thickness verification

The acceptance criteria in line with the section UG-27 as per ASME-BVPC Sect. VIII Div. 1. The actual thickness of the barrel must not be lower than the minimum calculated one, which is reliable for thickness lower than 1.5 times the inside radius or for internal pressure lower than 0.385SE.

8. References

Fundamentals of Machine Component Design – 5th Ed. – R.C. Juvinall, K.M. Marshek

Roarks' Formula for Stress and Strain – 8th Ed. – Warren C. Young, Richard G. Budynas, Ali M. Sadegh

Shingley's Mechanical Engineering Design – 10th Ed. – R. G. Budynas, J. K. Nisbet

ASME Boiler & Pressure Vessel Code VIII, Division 1, 2023 Edition

***SOP-477** - Bolted joint (without gasket) verification,*

***Mod.432** - Stud calculator*

***Mod.551** - Barrel and cover's bolted design*

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9. Appendix

9.1. Calculation Tool

The following instructions must be considered to carry out the calculation by means of the dedicated tool Mod.551:

1) Compile the input data.

Design condition		
T_{amb} [°C]	20.00	Ambient temperature
T_{design} [°C]	50.00	Design temperature
p_{design} [barg]	100.00	Design pressure
$p_{hydrotest}$ [barg]	150.00	Hydrotest pressure
Bolt		
E_b [GPa]	210.00	Young's modulus of bolt material
σ_{yb} [MPa]	724.00	Yield stress of bolt at ambient tempearture
σ'_{yb} [MPa]	724.00	Yield stress of bolt at design temperature
σ'_{ub} [MPa]	862.00	Ultimate stress of bolt at design temperature
α_b [mm]	1.20E-05	Bolt thermal expansion coefficient
d_n [mm]	80.00	Nut diameter
d_u [mm]	45.00	Bolt diameter, unthreaded part
L_u [mm]	110.00	Lenght of unthreaded part
d_t [mm]	52.00	Bolt diameter, threaded part
p [mm]	5.00	Pitch
s_w [mm]	0.00	Washer thickness
N_{bolts} [-]	20	Number of bolts
ξ [-]	80%	Tightening ratio, usually 70 - 80%
d_r [mm]	*47.31*	Tensile stress bolt diameter
L_t [mm]	*26.00*	Lenght of threaded part
A_u [mm^2]	*1590.43*	Area of unthreaded part
A_t [mm^2]	*1757.83*	Area of threaded part
L_e [mm]	*136.00*	Bolt effective total lenght
k_b [N/mm]	*2501353*	Bolt stiffness

Figure 9.1: General and bolt data.

Barrel and cover		
α [°]	30	Half apex angle of Rotscher pressure cone
ni [-]	0.30	Poisson's ratio
σ_{yf} [MPa]	450.00	Yield stress of barrel and cover at ambient temperature
σ'_{yf} [MPa]	450.00	Yield stress of barrel and cover at design temperature
σ'_{uf} [MPa]	690.00	Ultimate stress of barrel and cover at design temperature
E_f [GPa]	210.00	Young's modulus of barrel and cover material
d [mm]	56.00	Hole diameter
t_b [mm]	435.00	Barrel flange thickness
t_H [mm]	110.00	Cover flange thickness
L_h [mm]	70.00	Lenght of threaded part of the hole
α_f [1/°C]	1.70E-05	Flange thermal expansion coefficient
C [mm]	680.00	Bolt circle diameter
A [mm]	800.00	Outside diameter of the barrel / discharge casing
g_1 [mm]	135.00	Thickness of hub at back of flange (barrel)
g_0 [mm]	69.00	Thickness of hub at small end (barrel) - barrel thickness
h [mm]	125.00	Hub length
d_{int} [mm]	536.00	Barrel internal diameter
D_{cover} [mm]	309.00	Cover internal diameter
$D_{shaft,d}$ [mm]	115.00	Shaft diameter in design condition (minimum diameter at which design pressure is applied)
$D_{shaft,h}$ [mm]	80.00	Shaft diameter in hydrotest (minimum diameter at which hydrotest pressure is applied)
k_f [N/mm]	*10705241*	Flange stiffness
n_t [-]	*13*	Number of threads gripping the nut except one
F_{max} [N]	*2069092*	Maximum admissible load on thread

Figure 9.2: Barrel and cover data.

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Gasket		
type	Spiralwound	Gasket type
manufacturer	Flexitallic	Gasket manufacturer
name	Graphite filler and 321/1.4541	Gasket name
t_{gasket} [mm]	4.600	Gasket nominal thickness
T [°C]	23	Gasket characteristic temperature
t_f [mm]	4.00	Gasket operating thickness
D_{ext} [mm]	576.00	Gasket operating thickness
D_{int} [mm]	536.00	Gasket seat internal diameter
G [mm]	556.00	Diameter at location of gasket load reaction
δ_{Qa} [mm]	0.600	Gasket compression
$Q_{A\,max}$ [MPa]	74.20	Initial gasket pressure load
H_G [N]	1296121	Initial gasket load
$Q_{A\,max}$ [MPa]	29.38	Final gasket pressure load at design condition
H_G [N]	513200	Final gasket load at design condition
$Q_{A\,max}$ [MPa]	15.63	Final gasket pressure load at hydrotest condition
H_G [N]	272980	Final gasket load at hydrotest condition

Figure 9.3: Gasket data.

2) Verify that the results are satisfied according to the acceptance criteria in all the load conditions for all the sections.

Bolted joint verification			
σ_b/σ_{yb} [-]	88.51%	92.43%	(bolt stress ratio less than 90%)
F_i/F_0 [-]	21.84%	59.27%	(contact force reduction less than 60% of initial tightening)
$F_{b,tot}/F_{max}$ [-]	49.26%	51.44%	(bolt axial force and max admissible force for barrel thread less than 80%)

Figure 9.4: Acceptance criteria legend.

Bolting verification				
σ_b/σ_{yb} [-]	85.16%	88.75%	87.74%	(bolt stress ratio less than 90%)
F_i/F_0 [-]	27.60%	23.11%	41.40%	(contact force reduction less than 60% of initial tightening)
$F_{b,tot}/F_{max}$ [-]	47.39%	49.39%	48.83%	(bolt axial force and max admissible force for barrel thread less than 80%)

Figure 9.5: Acceptance criteria for bolted joint.

ASME VIII Div. 1 - Appendix Y - Acceptance criteria (Y-7)		
S_f [MPa]	197.14	Allowable design stress for material of flange at design temperature (operating condition) or atmospheric temperature (gasket seating)
S_B [MPa]	724.00	Allowable bolt stress at design temperature
σ_b/S_B [-]	0.33	Ratio between operating bolt stress and allowable bolt stress at design temperature (Y-7a)
$S_{H\,I}/S_f$ [-]	0.02	Ratio between longitudinal hub stress and allowable design stress for flange at design temperature (Y-7b)
$S_{R\,I}/S_f$ [-]	0.00	Ratio between radial stress in Flange I at Bolt Circle and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,I}/S_f$ [-]	0.01	Ratio between radial stress in Flange I at Inside Diameter and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,II}/S_f$ [-]	0.78	Ratio between radial stress in Flange II at Bolt Circle and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,II}/S_f$ [-]	0.17	Ratio between radial stress in Flange II at Diameter B1 and allowable design stress for flange at design temperature (Y-7c)
$S_{T\,I}/S_f$ [-]	0.02	Ratio between tangential stress in Flange I at Inside Diameter and allowable design stress for flange at design temperature (Y-7d)
$S_{T\,II}/S_f$ [-]	0.13	Ratio between tangential stress in Flange II at Diameter B1 and allowable design stress for flange at design temperature (Y-7d)
$(S_{H\,I} + S_{R\,I})/2/S_f$ [-]	0.01	Ratio between average radial & longitudinal stress in Flange I and allowable design stress for flange at design temperature (Y-7e)
$(S_{H\,I} + S_{T\,I})/2/S_f$ [-]	0.11	Ratio between average radial & tangential stress in Flange I and allowable design stress for flange at design temperature (Y-7e)
$S_{R\,II}/S_f$ [-]	0.42	Ratio between radial stress in Flange II at Center of Flange and allowable design stress for flange at design temperature (Y-7f)
$S_{T\,II}/S_f$ [-]	0.42	Ratio between tangential stress in Flange II at Center of Flange and allowable design stress for flange at design temperature (Y-7d)

ASME VIII Div. 1 - Appendix II - Flange rigidity factors (Table 2-14)		
M_o [Nm]	173955	Total moment for integral-type flanges and optional type flanges designed as integral-type flanges
e	0.00	Factor for integral type flange
d	462173	Factor for integral type flange
L	3.74	Factor for integral type flange
J	0.01	Rigidity factor for integral-type flanges and optional type flanges designed as integral-type flanges

Figure 9.6: Acceptance criteria as per ASME VIII Div. 1 Appendix Y and Appendix II.

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ASME VIII Div. 1 - UG-27 - Thickness of shells under internal pressure		
t_{barrel} [mm]	69.00	Barrel thickness
E	1.00	Joint efficiency (in case of casting, equal to 1)
SF	3.00	Safety factor for barrel thickness (minimum 2.5)
p [barg]	100.00	Internal design pressure
R [mm]	268.00	Inside radius of the shell
S [MPa]	450.00	Maximum allowable stress value
t_{min} [mm]	18.11	Minimum required thickness at hydrostatic pressure
t/t_{min} [-]	0.26	Ratio between barrel minimum required and actual thicknesses

Figure 9.7: Acceptance criteria as per ASME VIII Div. 1 Section UG-27.

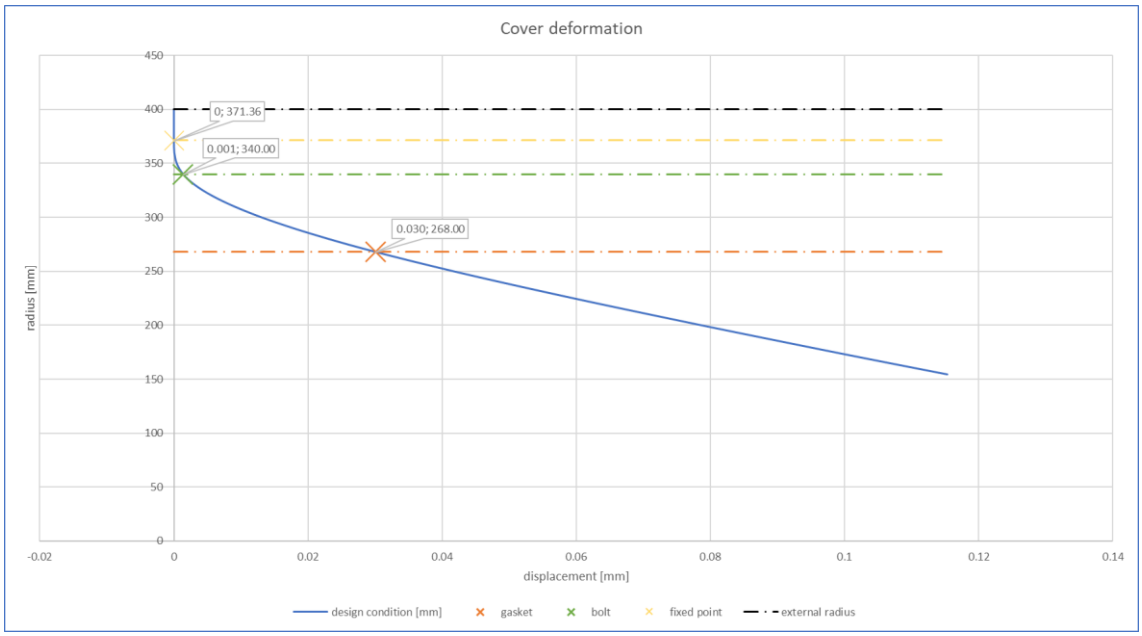

Figure 9.8: Calculation of cover deformation.

3) If the results are not acceptable, the input data must be modified and the verification performed again. If not possible, further analysis must be done.

9.2. *Case study – Job P30640.002 / Item 20157*

The case study in subject is related to BB5 pump model MESB200.6, installed in West Island of Zakum Field. The reference operating and design data are the following (refer to engineering documentation for further details):

- Operating fluid: Treated sea water
- Relative density: 1.033 kg/dm3
- Operating flow: 529.9 m3/h
- Differential head: 2002 m
- Speed: 1485 rpm
- MAWP: 277.7 barg
- Design temperature: 48.9°C
- Hydrotest pressure: 416.6 barg

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- Bolts type: N. 20 x M52
- Barrel and cover material: ASTM A955 6A (yield stress 450 MPa)
- Bolts material: ASTM A193 B7 (yield stress 724 MPa)

The results calculated with the spreadsheet are reported in *Figure 9.9* and *Figure 9.10*.

Bolted joint verification					
σ_b/σ_{yb} [-]	!	88.81%	✗	92.43%	(bolt stress ratio less than 90%)
F_J/F_b [-]	✓	21.76%	!	59.06%	(contact force reduction less than 60% of initial tightening)
$F_{b,tot}/F_{max}$ [-]	✓	49.42%	✓	51.44%	(bolt axial force and max admissible force for barrel thread less than 80%)

Figure 9.9: Results for bolted joint.

ASME VIII Div. 1 - Appendix Y - Acceptance criteria (Y-7)			
S_f [MPa]		197.14	Allowable design stress for material of flange at design temperature (operating condition) or atmospheric temperature (gasket seating)
S_B [MPa]		724.00	Allowable bolt stress at design temperature
σ_b/S_B [-]	✓	0.92	Ratio between operating bolt stress and allowable bolt stress at design temperature (Y-7a)
$S_{H\,I}/S_f$ [-]	✓	0.05	Ratio between longitudinal hub stress and allowable design stress for flange at design temperature (Y-7b)
$S_{R\,I}/S_f$ [-]	✓	0.00	Ratio between radial stress in Flange I at Bolt Circle and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,I}/S_f$ [-]	✓	0.02	Ratio between radial stress in Flange I at Inside Diameter and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,II}/S_f$ [-]	✗	2.16	Ratio between radial stress in Flange II at Bolt Circle and allowable design stress for flange at design temperature (Y-7c)
$S_{R\,II}/S_f$ [-]	✓	0.47	Ratio between radial stress in Flange II at Diameter B1 and allowable design stress for flange at design temperature (Y-7c)
$S_{T\,I}/S_f$ [-]	✓	0.06	Ratio between tangential stress in Flange I at Inside Diameter and allowable design stress for flange at design temperature (Y-7d)
$S_{T\,II}/S_f$ [-]	✓	0.36	Ratio between tangential stress in Flange II at Diameter B1 and allowable design stress for flange at design temperature (Y-7d)
$(S_{H\,I} + S_{R\,I})/2/S_f$ [-]	✓	0.03	Ratio between average radial & longitudinal stress in Flange I and allowable design stress for flange at design temperature (Y-7e)
$(S_{H\,I} + S_{T\,I})/2/S_f$ [-]	✓	0.30	Ratio between average radial & tangential stress in Flange I and allowable design stress for flange at design temperature (Y-7e)
$S_{R\,II}/S_f$ [-]	✗	1.16	Ratio between radial stress in Flange II at Center of Flange and allowable design stress for flange at design temperature (Y-7f)
$S_{T\,II}/S_f$ [-]	✗	1.16	Ratio between tangential stress in Flange II at Center of Flange and allowable design stress for flange at design temperature (Y-7d)

Figure 9.10 Results as per ASME VIII Div. 1 - Appendix Y.

The design solution did not meet the acceptance criteria as indicated in the previous sections.

For this reason, detailed FEM analysis through the software Marc Mentat (version 2024.2) was performed to validate the design. The results are presented in the following: the stress levels on casings and bolts are acceptable as shown in *Figure 9.11* and *Figure 9.12*.

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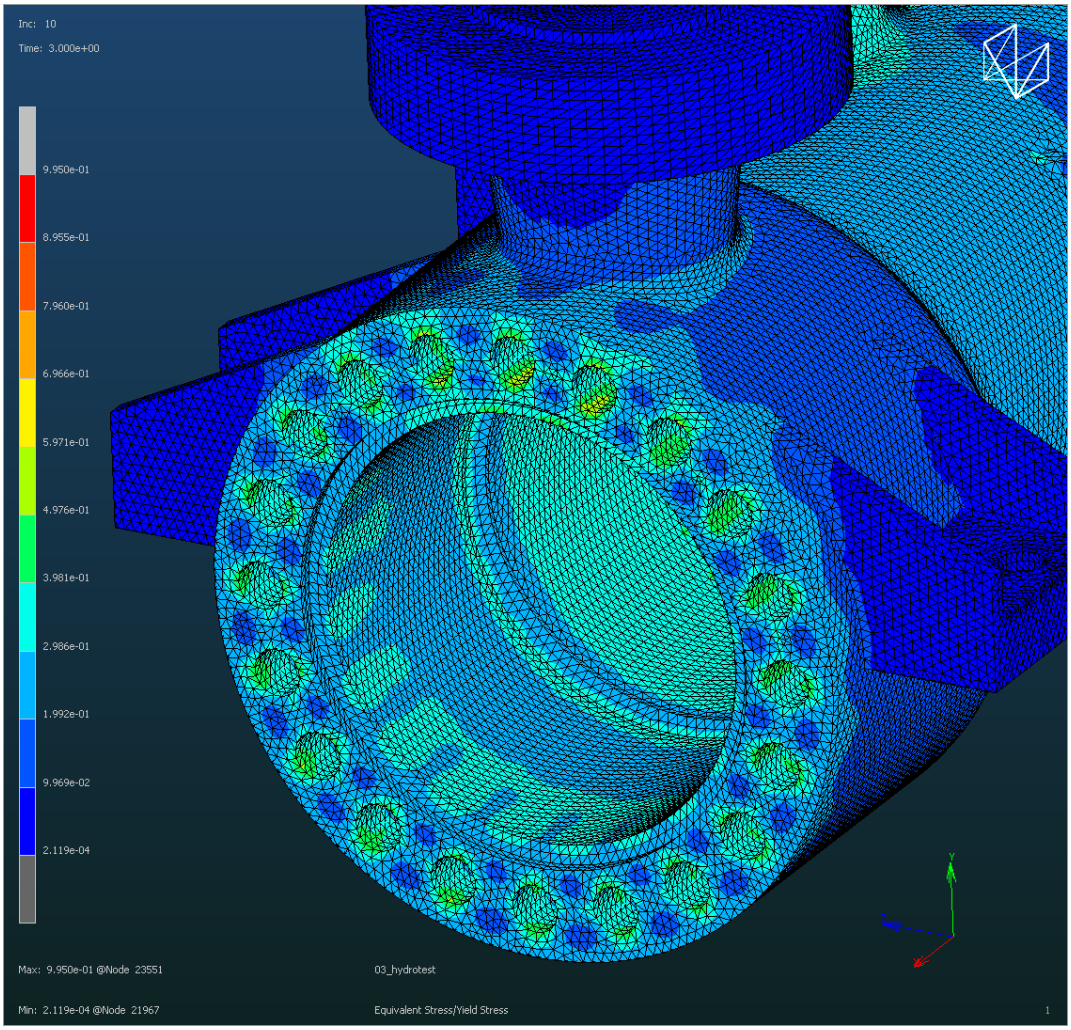

Figure 9.11: Ratio between Von Mises equivalent and allowable stresses in hydrotest: barrel casing.

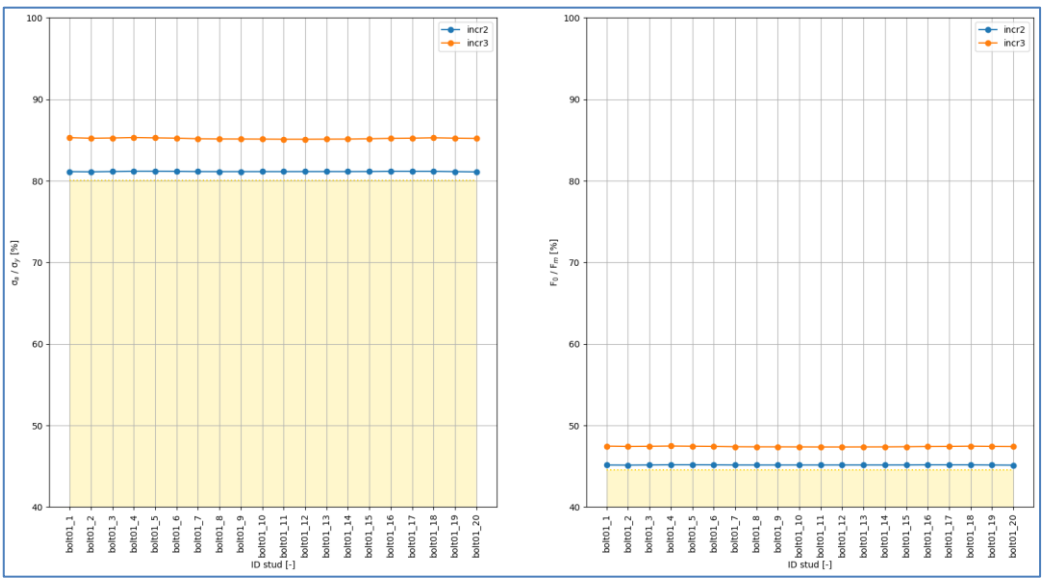

Figure 9.12: Bolts stress level at design pressure (incr2) and hydrotest (incr3).

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